ANAYA SHENDE

Computer Science & Engineering Undergraduate · AI/ML & Data Science

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ABOUT

Dedicated CS undergraduate with hands-on expertise in Python, AI/ML, and end-to-end system development. Self-directed ML learner who independently built RAG pipelines, satellite imagery enhancement systems, and full-stack ML web apps outside of coursework. IEEE member with Red Hat certifications. Actively seeking software engineering or ML internship roles at Google and Microsoft to contribute to impactful, large-scale technical projects.

EDUCATION

B.Tech — Computer Science & Engineering 2024 – 2028

Symbiosis Institute of Technology, Nagpur · **CGPA: 7.0**

Class 12 — PCMB 2024

Dinanath High School & Junior College, Dhantoli, Nagpur

TECHNICAL SKILLS

Languages	Python · C · JavaScript · SQL
AI / ML	LangChain · HuggingFace · ChromaDB · NumPy · scikit-learn · TensorFlow · RAG · RIFE / Optical Flow
CS Fundamentals	Data Structures & Algorithms · Operating Systems · DBMS · Theory of Computation · Embedded Systems
Tools & Platforms	Git · GitHub · VS Code · Flask · Streamlit · Google Cloud · AWS Open Data · Jira · WSL2

PROJECTS

RAG Chatbot — Document Q&A System **AI / ML**

Problem *Manually searching large document collections is time-consuming and imprecise, with no way to ask natural-language questions and get contextually accurate answers.*

Solution Built a Retrieval-Augmented Generation pipeline that embeds documents into a ChromaDB vector store using HuggingFace sentence-transformers. On query, the system retrieves top-k relevant chunks and passes them to Groq LLM (LLaMA 3.3 70B) via LangChain, generating grounded responses. Deployed with a Flask backend and interactive HTML/CSS/JS frontend.

[Python] [LangChain] [ChromaDB] [HuggingFace] [Groq API] [Flask] [RAG]

Used Car Price Predictor — ML Web App **ML · Flask**

Problem *No reliable tool existed to estimate used car resale prices from Quikr listings without manual market research.*

Solution Cleaned and pre-processed raw Quikr dataset (year, kms, fuel type, brand extraction). Built a scikit-learn pipeline with OneHotEncoding and Linear Regression, serialised as a .pkl model. Deployed via Flask with a dynamic HTML/JS frontend that populates dropdowns from live CSV data and returns real-time price predictions via JSON.

[Python] [scikit-learn] [Linear Regression] [Flask] [pandas] [Pickle] [JavaScript]

Breast Cancer Classification — Neural Network **Deep Learning**

Problem *Manual histopathology diagnosis is slow and prone to human error; a reliable automated classifier is needed.*

Solution Built a TensorFlow/Keras sequential neural network (30 features → Dense(20, ReLU) → Dense(2, Sigmoid)) trained on the sklearn breast cancer dataset. Applied StandardScaler for feature normalisation, achieved high validation accuracy, and implemented a real-time single-sample prediction interface.

[Python] [TensorFlow] [Keras] [scikit-learn] [NumPy] [StandardScaler]

Diabetes Prediction — SVM Classifier ML · Healthcare

Problem *Early diabetes detection from clinical parameters (glucose, BMI, insulin, etc.) requires a reliable, interpretable ML model.*

Solution Trained a Support Vector Machine (linear kernel) on the PIMA Indians Diabetes dataset with StandardScaler pre-processing and stratified train/test split. Achieved robust accuracy on held-out data with a real-time prediction interface for individual patient records.

[Python] [scikit-learn] [SVM] [pandas] [NumPy] [StandardScaler]

University Management System DBMS

Problem *Managing student records, faculty assignments, course enrollments, and reporting was scattered and error-prone without a centralised, structured database system.*

Solution Designed a fully normalised relational database schema (3NF) covering students, faculty, courses, and enrollments. Implemented optimised SQL queries for CRUD operations, multi-table joins, and automated reporting to eliminate redundancy and enforce data integrity.

[MySQL] [Relational Modelling] [SQL Joins] [3NF Normalisation] [PL/SQL]

Gesture-Controlled Music Player Computer Vision · Browser

Problem *Traditional media controls require physical interaction; a hands-free browser-based controller using real-time gesture recognition was needed.*

Solution Built a fully browser-based application using MediaPipe Hands for real-time hand landmark detection and the HTML5 Audio API for media control. Gestures map to play/pause, volume, and track navigation — zero backend, zero installs. Inspired by Charli XCX's 'Track 10'.

[JavaScript] [MediaPipe Hands] [HTML5 Audio API] [CSS] [Browser-based]

GaN HEMT RF Parameter Prediction — ML Research Research · Semiconductors

Problem *Predicting RF and power device parameters from GaN HEMT fabrication inputs requires expensive simulation; an ML surrogate model can accelerate design.*

Solution Independently curated a dataset from ~16 research papers (~25 target) covering GaN HEMT fabrication parameters. Training Neural Network, Random Forest, and Genetic Algorithm models to predict device RF/power outputs. Self-directed research project targeting publication-quality results.

[Python] [scikit-learn] [Neural Network] [Random Forest] [Genetic Algorithm] [Pandas]

PATENT

System and Method for Processor-Based Optimization of Gallium Nitride High Electron Mobility Transistor Structures 2025

Applicant: Symbiosis International (Deemed University), Pune

Inventors: Dr. Mohankumar N, Anaya Shende, Arya Khandekar, Bhumika Arora

IPC Classification: G06N 3/08 · G06N 3/04 · G06F 30/27 · G06N 20/00 · G06N 3/0475

Abstract: *Processor-based system and method for ML-driven optimization of GaN HEMT device structures, leveraging neural networks and evolutionary algorithms to predict and optimize RF/power device parameters from fabrication inputs — reducing dependency on expensive TCAD simulation.*

CERTIFICATIONS & ACHIEVEMENTS

Python Programming with Red Hat (AD141) · Red Hat — 32 Credit Hours · Hands-on Python in RHEL environments Mar 2025

Red Hat System Administration I (RH124) · Red Hat — 40 Credit Hours · Foundational Linux/RHEL administration Mar 2026

Data Analytics Certification · Infosys — Data analysis, Python workflows, analytical problem-solving 2025

Bharatiya Antariksh Hackathon — Idea Submission · ISRO × Hack2Skill — Code: 2025H2S06BAH25-P06975 · Research-oriented space tech solution 2025

Google Cloud Agentic AI Day — Participant · Google Cloud × Hack2Skill — AI-driven intelligent agent system · ID: 2025H2S06AID-I02178 2025

IEEE Student Member · IEEE — Technical events, research discussions, engineering community 2024 – Present

Debate Competition · The Orator Club, IIIT Nagpur — Communication, critical thinking, competitive public speaking 2024

LANGUAGES

English (Professional) · **Hindi** (Professional) · **Marathi** (Native) · **German** (Beginner)